Lakshya Jindal

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Education

International Institute of Information Technology, Hyderabad Aug 2023 – Present
B.Tech + M.S. (By Research) in Electronics and Communication Engineering *Hyderabad, India*

Hiranandani Foundation School Thane June 2021 – June 2023
ISC, PCM and Computer Applications - 94.25% *Mumbai, India*

Experience

Undergraduate Researcher | Robotics Research Centre, IIIT Hyderabad April 2025 – Present
Research Assistant *On-site*

- Designed and implemented scalable motion planning and perception algorithms in C++ and Python, applying linear algebra, probability, and numerical optimization to high-dimensional configuration spaces.
- Architected modular, low-latency software pipelines for large-scale simulation and algorithm evaluation on Linux, emphasizing production-quality engineering, reproducibility, and automated benchmarking.

Publications

Towards Kinematic Actionable Infeasibility Detection in Motion Planning | *IEEE RA-L (Accepted)* | arXiv

- Co-authored research introducing a geometry-driven framework for certifying path non-existence in high-DOF kinematic motion planning, leveraging GPU-accelerated simplicial reconstruction.

Projects

Scaling Motion Planning Infeasibility Proofs | *C++, Python, OMPL, GUDHI, FCL* | GitHub

- Implemented the asymptotically complete motion planner from Li & Dantam (IEEE RA-L 2023), running PRM, Sample-Driven Connectivity Learning (SDCL), and proof construction across concurrent threads.
- Engineered parallel codimension-one manifold tracing via Coxeter triangulation with concurrent hash tables, integrating local elastic vertex updates and recursive simplex bisection to certify $\mathcal{C}_{obs}$ containment.

Target-Less LiDAR-Camera Calibration (MDPCalib) | *Python, Computer Vision, Sensor Fusion, Optimization* | Paper

- Implemented target-less extrinsic calibration from scratch based on the MDPCalib framework, aligning visual and LiDAR odometry trajectories for initialization-free coarse sensor registration.
- Developed the complete pipeline including feature extraction, deep 2D-pixel-to-3D-point correspondence matching, and Cauchy-robustified non-linear joint optimization for accurate multi-modal alignment.

Multi-Modal Batch-MPC for Autonomous Driving | *C++, Python, Eigen, Optimal Control, MPC* | Paper

- Implemented a real-time batch non-holonomic trajectory optimizer based on Adajania et al. (IEEE RA-L 2022) to generate parallel collision-free candidate trajectories for multi-modal highway driving maneuvers.
- Formulated a linearization-free multi-convex relaxation of kinematic, acceleration, and collision constraints solved via Alternating Minimization, reducing parallel iterations to fast vectorized matrix-vector products.

Real-Time Object Detection in Gazebo | *ROS 1, Python, PyTorch, YOLOv5, OpenCV, Gazebo* | GitHub

- Developed a modular ROS 1 perception package for real-time 2D object detection in Gazebo simulation, subscribing to simulated RGB camera streams via `cv_bridge` for low-latency frame processing.
- Integrated a pre-trained YOLOv5 inference pipeline with custom ROS publishers to broadcast bounding-box coordinates and class confidence telemetry for downstream autonomous navigation tasks.

Technical Skills

Languages: C/C++, Python (NumPy, SciPy, Pandas), MATLAB, Bash/Shell, SQL
Robotics & Control: ROS 1/2, Gazebo, OMPL, FCL, Motion Planning, MPC, Trajectory Optimization, Sensor Fusion, Kinematics
Perception & AI: OpenCV, PyTorch, YOLO, Point Cloud Processing (PCL), Computer Vision, Deep Learning
Libraries & Tools: Eigen, NLopt, GUDHI, Git, Linux, CMake, Multi-threading / Parallel Computing

Achievements/Extracurriculars

- Selected among the **Top 45 Teams** in the **Adobe University Hackathon 2026**.
- Secured **3rd Place** in **Megathon 2025**, building a Multi-Modal AI Solution for Estimating Vehicle Damage Claims.
- Achieved a peak rating of **1269** on **Codeforces**, solving 200+ algorithms and data structures problems.
- Secured **AIR 4391** in JEE Mains & **AIR 10931** in JEE Advanced; Cleared **KVPY 2022** & **Pre-RMO 2019**.

Relevant Coursework

Robotics: Planning and Navigation | Computer Vision | Signal Processing | Probability & Random Processes | Data Structures and Algorithms | Statistical Methods in AI | Linear Algebra | Operating Systems | Control Systems